Lecture 47-48: Generating Functions, Generalized Inclusion-Exclusion Principle

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Outline

- Generating functions: An introduction
- Useful facts about formal power series
  - Some basic facts
  - Extended binomial theorem
  - Some useful generating functions
- Solving recurrence relations by generating functions
- Generalized Inclusion-exclusion principle
  - Formulation and proof of the principle
  - Applications
    - Sieves of Eratosthenes
    - Number of onto functions
    - Derangements

Generating Functions

A way of encoding an infinite sequence of numbers $\{a_n\}$
by treating them as the coefficients of a formal power series.

The generating function (series) of the sequence

Example. $\sum_{k=0}^{\infty} x^k$ is the generating function for the sequence $\{1\}$.

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Often expressed in closed form (rather than as a series),
by some expression involving operations defined for formal series.

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First introduced by Abraham de Moivre in 1730.

Abraham de Moivre (1667 -1754): French Mathematician

Generating Functions

The *generating function* for the infinite sequence $a_0, a_1, \ldots, a_k, \ldots$ of real numbers is the infinite formal power series

$$G(x) = a_0 + a_1x + a_2x^2 + \ldots + a_kx^k + \ldots = \sum_{k=0}^{\infty} a_kx^k.$$

The generating function for the sequence $\{1\}$ is $\sum_{k=0}^{\infty} x^k$.

The generating function for the sequence $\{k+1\}$ is $\sum_{k=0}^{\infty} (k+1)x^k$.

The generating function for the sequence $\{2^k\}$ is $\sum_{k=0}^{\infty} 2^k x^k$.

Generating Functions

The *generating function* for the finite sequence $a_0, a_1, \ldots, a_n$ of real numbers is a polynomial of degree n,

$$G(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n = \sum_{k=0}^{n} a_k x^k.$$

The generating function for the sequence {1, 1, 1, 1} is

$$G(x) = \sum_{k=0}^{3} x^k = (x^4 - 1) / (x - 1).$$

Let $m \in \mathbb{Z}_+$. The generating function for the sequence

$\{C(m, 0), \ldots, C(m, m)\}$ is $G(x) = \sum_{k=0}^{m} C(m,k) x^k = (1+x)^m$.

Counting By Generating Functions

使用1元、2元人民币凑n元零钱的方法数

The coefficient of x^n in

$$(1 + x^1 + x^2 + \ldots)(1 + x^2 + x^4 + \ldots) = \frac{1}{1-x}\frac{1}{1-x^2}$$

$$\frac{1}{1-x}\frac{1}{1-x^2} = \frac{1}{(1-x)^2(1+x)} = \frac{1}{4}\frac{3-x}{(1-x)^2} + \frac{1}{4}\frac{1}{1+x}$$

Counting By Generating Functions

使用1元、2元人民币凑n元零钱的方法数

$$\frac{1}{4}\frac{3-x}{(1-x)^2}+\frac{1}{4}\frac{1}{1+x} = \frac{1}{4}(3-x)\sum_{n=0}^{\infty}(n+1)x^n+\frac{1}{4}\sum_{n=0}^{\infty}(-1)^n x^n$$

$$= \frac{1}{4}\sum_{n=0}^{\infty}3(n+1)x^n-\frac{1}{4}\sum_{n=0}^{\infty}(n+1)x^{n+1}+\frac{1}{4}\sum_{n=0}^{\infty}(-1)^n x^n$$

$$= \frac{1}{4}\sum_{n=0}^{\infty}3(n+1)x^n-\frac{1}{4}\sum_{n=0}^{\infty}nx^n+\frac{1}{4}\sum_{n=0}^{\infty}(-1)^n x^n$$

$$= \frac{1}{4}\sum_{n=0}^{\infty}\left(2n+3+(-1)^n\right)x^n$$

Counting By Generating Functions

使用1元、2元人民币凑n元零钱的方法数

The coefficient of x^n in

$$(1 + x^1 + x^2 + \ldots)\,(1 + x^2 + x^4 + \ldots)$$

$$\boxed{\frac{1}{4}\left(2n + 3 + (-1)^n\right)}$$

Counting By Generating Functions

How many ways to distribute 8 identical cookies to 3 distinct children, with each child receiving at least 2 and at most 4 cookies.

Counting By Generating Functions

How many ways to distribute 8 identical cookies to 3 distinct children, with each child receiving at least 2 and at most 4 cookies.

The coefficient of x^8 in

$$(x^2 + x^3 + x^4)\,(x^2 + x^3 + x^4)\,(x^2 + x^3 + x^4)$$

Counting By Generating Functions

How many ways to distribute 8 identical cookies to 3 distinct children, with each child receiving at least 2 and at most 4 cookies.

The coefficient of x^8 in

$$(x^2 + x^3 + x^4)\,(x^2 + x^3 + x^4)\,(x^2 + x^3 + x^4)$$

$$x^{12} + 3x^{11} + 6x^{10} + 7x^9 + 6x^8 + \cdots$$

Counting By Generating Functions

How many ways to distribute 8 identical cookies to 3 distinct children, with each child receiving at least 2 and at most 4 cookies.

The coefficient of x^8 in

$$(x^2 + x^3 + x^4)\,(x^2 + x^3 + x^4)\,(x^2 + x^3 + x^4)$$

$$x^{12} + 3x^{11} + 6x^{10} + 7x^9 + 6x^8 + \cdots$$

2, 2, 4   2, 4, 2   4, 2, 2   2, 3, 3   3, 2, 3   3, 3, 2

Solving Recurrence Relations By Generating Functions

Solve $F_n = F_{n\text{-}1} + F_{n\text{-}2}$, $F_0 = 0$, $F_1 = 1$.

Let $G(x) = \sum_{n=0}^{\infty} F_n x^n$.

Then $G(x) = F_0 + F_1 x + \sum_{n=2}^{\infty} F_n x^n = x + \sum_{n=2}^{\infty} (F_{n-1} + F_{n-2}) x^n$

$$= x + \sum_{n=2}^{\infty} F_{n-1} x^n + \sum_{n=2}^{\infty} F_{n-2} x^n \;= x + x \sum_{n=0}^{\infty} F_n x^n + x^2 \sum_{n=0}^{\infty} F_n x^n$$

$$= x + xG(x) + x^2 G(x)$$

Solving Recurrence Relations By Generating Functions

Solve $F_n = F_{n\text{-}1} + F_{n\text{-}2}$, $F_0 = 0$, $F_1 = 1$.

Let $G(x) = \sum_{n=0}^{\infty} F_n x^n$.

$$G(x) = x + xG(x) + x^2G(x)$$

$$\begin{aligned} G(x) &= \frac{x}{1 - x - x^2} \\ &= -x\frac{1}{\left(x + \frac{1+\sqrt{5}}{2}\right)\left(x + \frac{1-\sqrt{5}}{2}\right)} \end{aligned}$$

Solving Recurrence Relations By Generating Functions

Solve $F_n = F_{n-1} + F_{n-2}$, $F_0 = 0$, $F_1 = 1$.

Let $G(x) = \sum_{n=0}^{\infty} F_n x^n$.

$$G(x) = -x \frac{1}{\left(x + \frac{1+\sqrt{5}}{2}\right)\left(x + \frac{1-\sqrt{5}}{2}\right)}$$

$$= \frac{x}{\sqrt{5}} \left(\frac{1}{x + \frac{1+\sqrt{5}}{2}} - \frac{1}{x + \frac{1-\sqrt{5}}{2}} \right) = \frac{x}{\sqrt{5}} \left(-\frac{1-\sqrt{5}}{2} \frac{1}{1 - \frac{1-\sqrt{5}}{2}x} + \frac{1+\sqrt{5}}{2} \frac{1}{1 - \frac{1+\sqrt{5}}{2}x} \right)$$

Solving Recurrence Relations By Generating Functions

Solve $F_n = F_{n\text{-}1} + F_{n\text{-}2}$, $F_0 = 0$, $F_1 = 1$.

Let $G(x) = \sum_{n=0}^{\infty} F_n x^n$.

$$G(x) = \frac{x}{\sqrt{5}} \left(-\frac{1-\sqrt{5}}{2} \frac{1}{1 - \frac{1-\sqrt{5}}{2}x} + \frac{1+\sqrt{5}}{2} \frac{1}{1 - \frac{1+\sqrt{5}}{2}x} \right)$$

$$= \frac{x}{\sqrt{5}} \left(-\frac{1-\sqrt{5}}{2} \sum_{n=0}^{\infty} \left(\frac{1-\sqrt{5}}{2}x \right)^n + \frac{1+\sqrt{5}}{2} \sum_{n=0}^{\infty} \left(\frac{1+\sqrt{5}}{2}x \right)^n \right)$$

Solving Recurrence Relations By Generating Functions

Solve $F_n = F_{n\text{-}1} + F_{n\text{-}2}$, $F_0 = 0$, $F_1 = 1$.

Let $G(x) = \sum_{n=0}^{\infty} F_n x^n$.

$$
\begin{aligned}
G(x) &= \frac{x}{\sqrt{5}}\left(-\frac{1-\sqrt{5}}{2}\sum_{n=0}^{\infty}\left(\frac{1-\sqrt{5}}{2}x\right)^n + \frac{1+\sqrt{5}}{2}\sum_{n=0}^{\infty}\left(\frac{1+\sqrt{5}}{2}x\right)^n\right) \\
&= \frac{1}{\sqrt{5}}\left(-\sum_{n=0}^{\infty}\left(\frac{1-\sqrt{5}}{2}\right)^{n+1} x^{n+1} + \sum_{n=0}^{\infty}\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} x^{n+1}\right) \\
&= \frac{1}{\sqrt{5}}\sum_{n=0}^{\infty}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)x^n
\end{aligned}
$$

Solving Recurrence Relations By Generating Functions

Solve $F_n = F_{n-1} + F_{n-2}$, $F_0 = 0$, $F_1 = 1$.

Let $G(x) = \sum_{n=0}^{\infty} F_n x^n$.

$$G(x) = \frac{x}{\sqrt{5}}\left(-\frac{1-\sqrt{5}}{2}\sum_{n=0}^{\infty}\left(\frac{1-\sqrt{5}}{2}x\right)^n + \frac{1+\sqrt{5}}{2}\sum_{n=0}^{\infty}\left(\frac{1+\sqrt{5}}{2}x\right)^n\right)$$

$$F_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$= \frac{1}{\sqrt{5}}\sum_{n=0}^{\infty}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)x^n$$

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Some Basic Facts

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x} \text{ for } |x| < 1.$$

Let $a \neq 0$.

$$\sum_{k=0}^{\infty} a^k x^k = \frac{1}{1-ax} \text{ for } |ax| < 1 \text{ or equivalently } |x| < 1/|a|.$$

Some Basic Facts

Theorem. Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g(x) = \sum_{k=0}^{\infty} b_k x^k$. Then

$$f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k \text{ and}$$

$$f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} a_j b_{k-j} \right) x^k.$$

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$$f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} a_j b_{k-j} \right) x^k.$$

Example. Suppose $f(x) = 1 / (1 - x)^2$. Then

$$f(x) = \left(\sum_{k=0}^{\infty} x^k\right)^2 = \sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} 1 \cdot 1 \right) x^k = \sum_{k=0}^{\infty} (k+1) x^k.$$

Extended Binomial Theorem

Let $u \in \mathbb{R}$ and $k \in \mathbb{N}$. Then the *extended binomial coefficient*

$$\begin{pmatrix} u \\ k \end{pmatrix} = \begin{cases} u(u-1)\cdots(u-k+1)/k! & \text{if } k > 0, \\ 1 & \text{if } k = 0. \end{cases}$$

Example.

$$\begin{pmatrix} -2 \\ 3 \end{pmatrix} = -2(-2-1)(-2-3+1)/3! = -4$$

$$\begin{pmatrix} 1/2 \\ 3 \end{pmatrix} = 1/2(1/2-1)(1/2-3+1)/3! = 1/16$$

Extended Binomial Theorem

Example.

$$\binom{-n}{r} = \frac{(-n)(-n-1)\cdots(-n-r+1)}{r!}$$

$$= \frac{(-1)^r(n)(n+1)\cdots(n+r-1)}{r!}$$

$$= \frac{(-1)^r(n+r-1)!}{r!(n-1)!}$$

$$= (-1)^r\binom{n+r-1}{r}$$

Extended Binomial Theorem

Extended Binomial Theorem.

Let $x \in \mathbb{R}$ with $|x| < 1$ and $u \in \mathbb{R}$. Then

$$(1 + x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k \qquad \text{Note: } 0^0 = 1$$

Extended Binomial Theorem

Proof. Consider the Taylor expansion of $f(x) = (1+x)^u$ at 0.

$$f^{(k)}(x) = u(u-1)\cdots(u-k+1)(1+x)^{u-k}$$

$$f^{(k)}(0) = u(u-1)\cdots(u-k+1)$$

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = \sum_{k=0}^{\infty} \frac{u(u-1)\cdots(u-k+1)}{k!} x^k = \sum_{k=0}^{\infty} \binom{u}{k} x^k$$

$\displaystyle\sum_{k=0}^{\infty} \binom{u}{k} x^k$ converges if $|x| < 1$.

Extended Binomial Theorem

Proof (*continued*).

Let $g(x) = \sum_{k=0}^{\infty} \binom{u}{k} x^k$. Then $g'(x) = ug(x) / (1+x)$.

$$g'(x) = \sum_{k=0}^{\infty} \binom{u}{k+1} (k+1)x^k$$

$$\begin{aligned} ug(x)/(1+x) &= u\left(\sum_{k=0}^{\infty} \binom{u}{k} x^k\right)\left(\sum_{k=0}^{\infty} (-1)^k x^k\right) \\ &= \sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} u \binom{u}{j} (-1)^{k-j}\right) x^k \end{aligned}$$

Prove by an induction on *k*,

$$\forall k \in \mathbb{N} \ \binom{u}{k+1} (k+1) = \sum_{j=0}^{k} u \binom{u}{j} (-1)^{k-j}$$

Extended Binomial Theorem

Proof (*continued*).

Let $h(x) = g(x) / (1+x)^u$.

Then $h'(x) = g'(x)/(1+x)^u - ug(x)/(1+x)^{u+1} =$

$= (ug(x) / (1+x))/(1+x)^u - ug(x)/(1+x)^{u+1} = 0.$

Thus $h(x) = c$ for some $c \in \mathbb{R}$.

Moreover, we know that $h(0) = g(0) = 1$.

Thus $h(\mathrm{x}) = 1$, so $g(x) = (1+x)^u$.

Extended Binomial Theorem

Examples.

$$\frac{1}{(1+x)^n} = (1+x)^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} x^k = \sum_{k=0}^{\infty} (-1)^k C(n+k-1, k) x^k$$

$$\frac{1}{(1-x)^n} = (1-x)^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} (-x)^k = \sum_{k=0}^{\infty} C(n+k-1, k) x^k$$

Extended Binomial Theorem

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$$\frac{1}{(1-x)^n} = (1-x)^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} (-x)^k = \sum_{k=0}^{\infty} C(n+k-1,k)x^k$$

Select r objects of n different kinds,

with at least one object selected for each kind.

$$\Downarrow$$

The coefficient of x^r in $(x + x^2 + \cdots)^n$

$$(x + x^2 + \ldots)^n = \frac{x^n}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)x^{k+n}$$

$k + n = r \Rightarrow$ the coefficient of x^r is $C(r\text{-}1, r\text{-}n) = C(r\text{-}1, n\text{-}1)$

Some Useful Generating Functions

TABLE 1 **Useful Generating Functions.**

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^{n} C(n,k)x^k$ $= 1 + C(n,1)x + C(n,2)x^2 + \cdots + x^n$	$C(n,k)$
$(1+ax)^n = \sum_{k=0}^{n} C(n,k)a^k x^k$ $= 1 + C(n,1)ax + C(n,2)a^2x^2 + \cdots + a^n x^n$	$C(n,k)a^k$
$(1+x^r)^n = \sum_{k=0}^{n} C(n,k)x^{rk}$ $= 1 + C(n,1)x^r + C(n,2)x^{2r} + \cdots + x^{rn}$	$C(n,k/r)$ if $r \mid k$; 0 otherwise
$\dfrac{1-x^{n+1}}{1-x} = \sum_{k=0}^{n} x^k = 1 + x + x^2 + \cdots + x^n$	1 if $k \le n$; 0 otherwise

Some Useful Generating Functions

TABLE 1 **Useful Generating Functions.**

$G(x)$	a_k
$\dfrac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$	1
$\dfrac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \cdots$	a^k
$\dfrac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \cdots$	1 if $r \mid k$; 0 otherwise
$\dfrac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \cdots$	$k+1$
$\dfrac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)x^k = 1 + C(n,1)x + C(n+1,2)x^2 + \cdots$	$C(n+k-1, k) = C(n+k-1, n-1)$

Some Useful Generating Functions

TABLE 1 **Useful Generating Functions.**

$G(x)$	a_k
$\dfrac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)(-1)^k x^k = 1 - C(n,1)x + C(n+1,2)x^2 - \cdots$	$(-1)^k C(n+k-1,k) = (-1)^k C(n+k-1,n-1)$
$\dfrac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)a^k x^k = 1 + C(n,1)ax + C(n+1,2)a^2x^2 + \cdots$	$C(n+k-1,k)a^k = C(n+k-1,n-1)a^k$
$e^x = \sum_{k=0}^{\infty} \dfrac{x^k}{k!} = 1 + x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots$	$1/k!$
$\ln(1+x) = \sum_{k=1}^{\infty} \dfrac{(-1)^{k+1}}{k} x^k = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \dfrac{x^4}{4} + \cdots$	$(-1)^{k+1}/k$

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Solving Recurrence Relations By Generating Functions

C_n: Catalan numbers

Count the number of ways to parenthesize the product of $n+1$ numbers $x_0 \cdot x_1 \cdot \cdots \cdot x_n$

$$C_0 = 1,\ C_1 = 1$$

$$C_n = C_0\ C_{n\text{-}1} + C_1\ C_{n\text{-}2} + \cdots + C_{n\text{-}1}\ C_0 = \sum_{k=0}^{n-1} C_k C_{n\text{-}1\text{-}k}$$

Solving Recurrence Relations By Generating Functions

C_n: Catalan numbers

Count the number of ways to parenthesize the product of $n+1$ numbers $x_0 \cdot x_1 \cdot \cdots \cdot x_n$

$$C_0 = 1,$$

$$C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$$

Solving Recurrence Relations By Generating Functions

Let $G(x) = \sum_{n=0}^{\infty} C_n x^n$

Then
$$
\begin{aligned}
G(x) &= \sum_{n=0}^{\infty} C_n x^n \\
&= 1 + \sum_{n=1}^{\infty} \left(\sum_{k=0}^{n-1} C_k C_{n-1-k} \right) x^n \\
&= 1 + x \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} C_k C_{n-k} \right) x^n \\
&= 1 + x \left(\sum_{n=0}^{\infty} C_n x^n \right) \left(\sum_{n=0}^{\infty} C_n x^n \right) \\
&= 1 + xG^2(x).
\end{aligned}
$$

Solving Recurrence Relations By Generating Functions

$$\text{Let } G(x) = \sum_{n=0}^{\infty} C_n\, x^n$$

$$G(x) = 1 + xG^2(x)$$

$$xG^2(x) - G(x) + 1 = 0$$

$$G(x) = \frac{1 \pm \sqrt{1-4x}}{2x} \Rightarrow xG(x) = \frac{1 \pm \sqrt{1-4x}}{2}$$

$$0G(0) = \frac{1 \pm 1}{2} \Rightarrow G(x) = \frac{1 - \sqrt{1-4x}}{2x}$$

Solving Recurrence Relations By Generating Functions

Let $G(x) = \sum_{n=0}^{\infty} c_n x^n$

$$G(x) = \frac{1-\sqrt{1-4x}}{2x} = \frac{1}{2x} - \frac{1}{2x}\sum_{n=0}^{\infty}\binom{1/2}{n}(-4x)^n$$

$$= \frac{1}{2x} - \frac{1}{2x} - \frac{1}{2x}\sum_{n=1}^{\infty}\binom{1/2}{n}(-4x)^n$$

$$= -\frac{1}{2x}\sum_{n=1}^{\infty}\frac{\frac{1}{2}(\frac{1}{2}-1)\cdots(\frac{1}{2}-n+1)}{n!}(-4)^n x^n$$

Solving Recurrence Relations By Generating Functions

Let $G(x) = \sum_{n=0}^{\infty} c_n x^n$

$$G(x) = -\frac{1}{2x}\sum_{n=1}^{\infty}\frac{\frac{1}{2}(\frac{1}{2}-1)\cdots(\frac{1}{2}-n+1)}{n!}(-4)^n x^n$$

$$=\sum_{n=1}^{\infty}\frac{1\cdot 3\cdot \cdots \cdot (2n-3)}{n!}2^{n-1}x^{n-1}$$

$$=\sum_{n=1}^{\infty}\frac{(2n-2)!}{n!(n-1)!}x^{n-1}=\sum_{n=0}^{\infty}\frac{1}{n+1}\frac{(2n)!}{n!n!}x^n$$

Solving Recurrence Relations By Generating Functions

Let $G(x) = \sum_{n=0}^{\infty} C_n x^n$

$$G(x) = -\frac{1}{2x}\sum_{n=1}^{\infty}\frac{\frac{1}{2}(\frac{1}{2}-1)\cdots(\frac{1}{2}-n+1)}{n!}(-4)^n x^n$$

$$C_n = \frac{1}{n+1}\binom{2n}{n}$$

$$= \sum_{n=1}^{\infty}\frac{(2n-2)!}{n!(n-1)!}x^{n-1} = \sum_{n=0}^{\infty}\frac{1}{n+1}\frac{(2n)!}{n!n!}x^n$$

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Generalized Inclusion-Exclusion Principle

The inclusion-exclusion principle

$$|A \cup B| = |A| + |B| - |A \cap B|$$

How many positive integers $\leq$ 1000 are divisible by 7 or 11 ?

A: positive integers $\leq$ 1000 divisible by 7

B: positive integers $\leq$ 1000 divisible by 11

$$|A| = \lfloor 1000/7 \rfloor = 142,\ |B| = \lfloor 1000/11 \rfloor = 90$$

$$|A \cap B| = \lfloor 1000/77 \rfloor = 12.$$

Generalized Inclusion-Exclusion Principle

The inclusion-exclusion principle

$|A \cup B| = |A| + |B| - |A \cap B|$

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A: positive integers $\leq$ 1000 divisible by 7

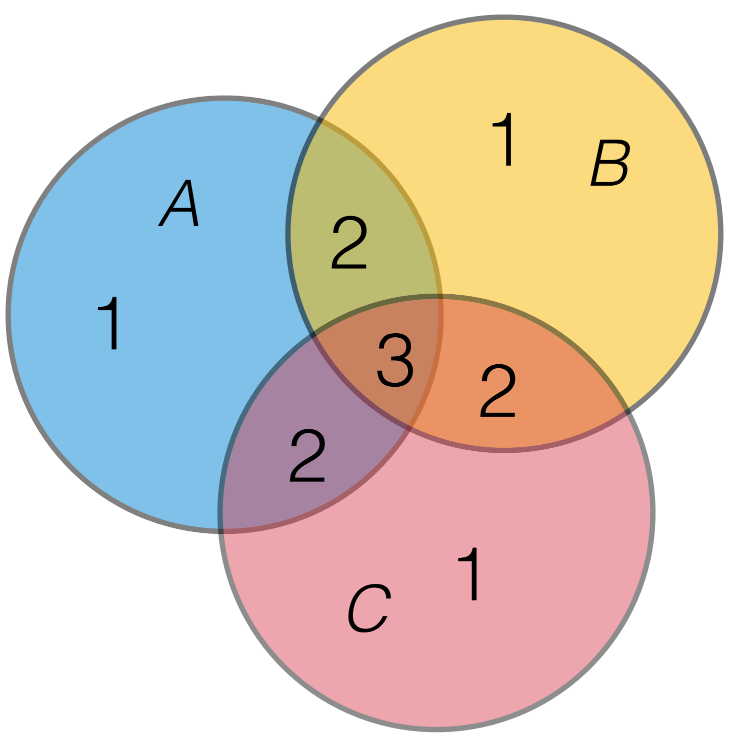
B: positive integers $\leq$ 1000 divisible by 11

$$|A| = \lfloor 1000/7 \rfloor = 142,\ |B| = \lfloor 1000/11 \rfloor = 90$$
$$|A \cap B| = \lfloor 1000/77 \rfloor = 12.$$

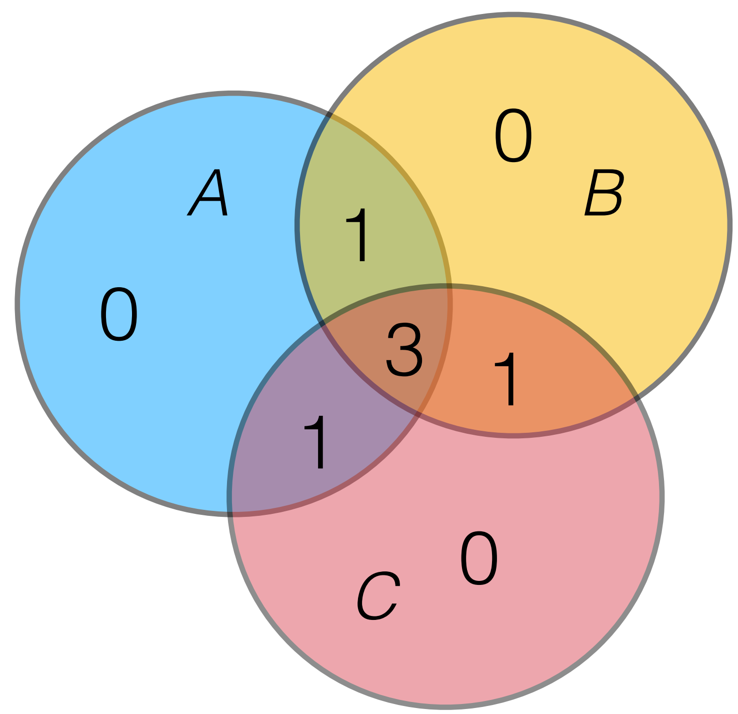
$$|A \cup B| = |A| + |B| - |A \cap B| = 142 + 90 - 12 = 220.$$

Generalized Inclusion-Exclusion Principle

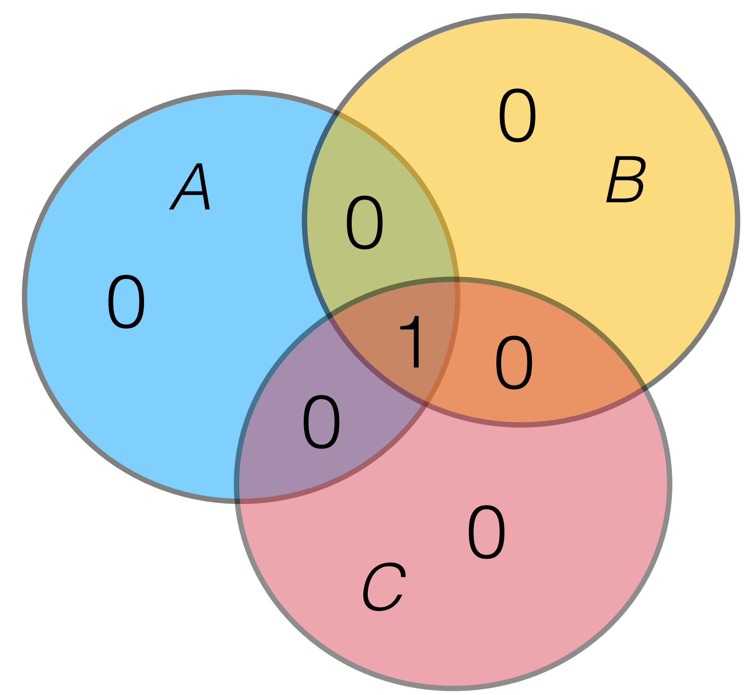
$$|A \cup B \cup C| =$$
$$|A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$



$|A| + |B| + |C|$



$|A \cap B| + |B \cap C| + |A \cap C|$



$|A \cap B \cap C|$

Generalized Inclusion-Exclusion Principle

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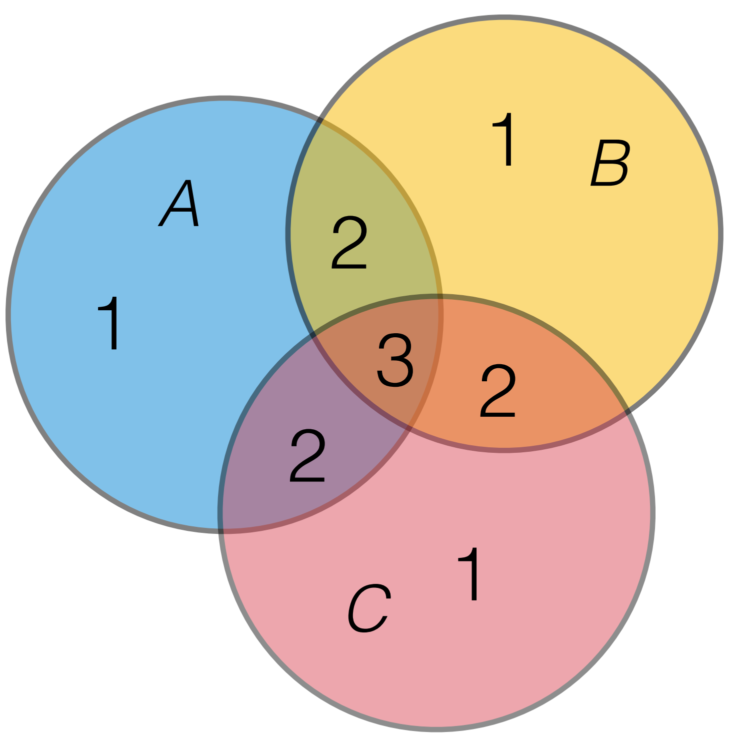
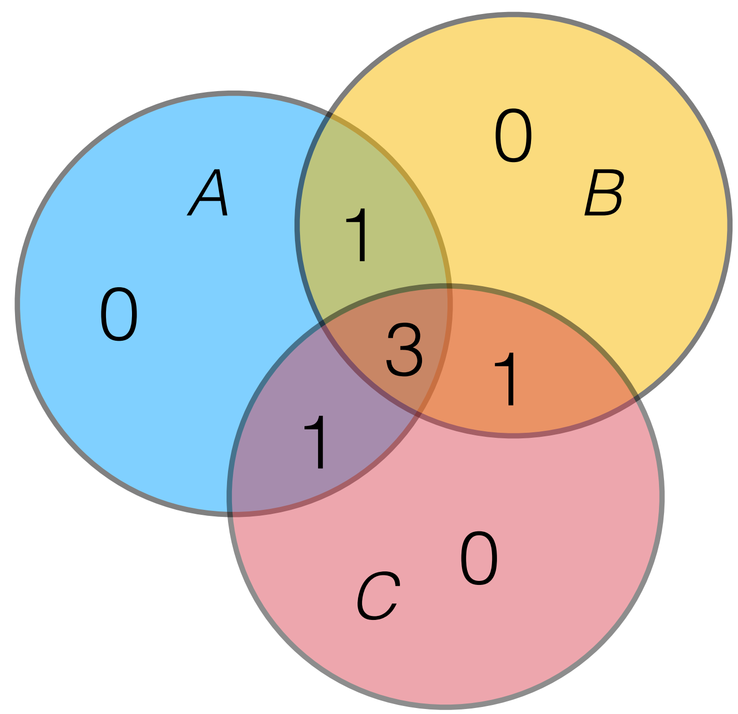


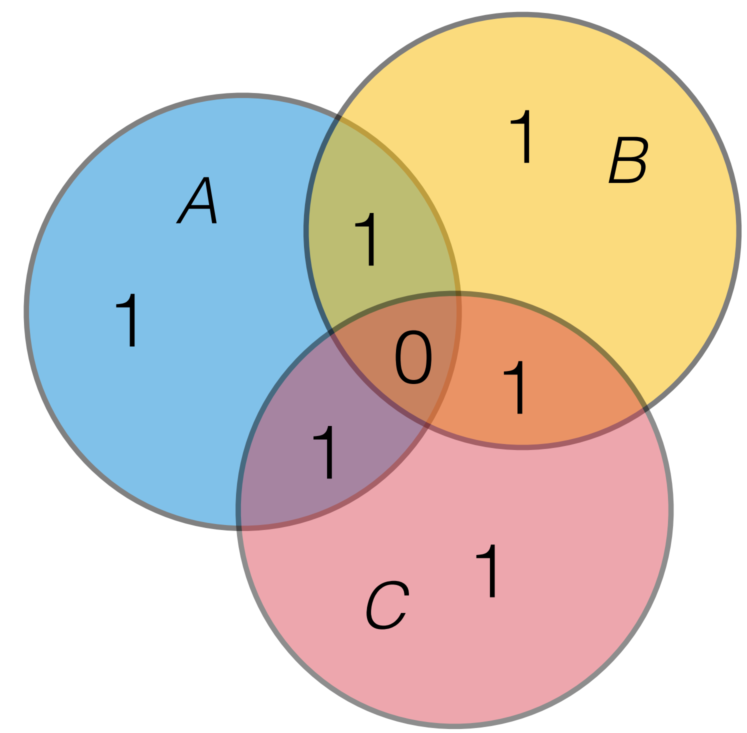
$|A| + |B| + |C|$

-



$|A \cap B| + |B \cap C| + |A \cap C|$

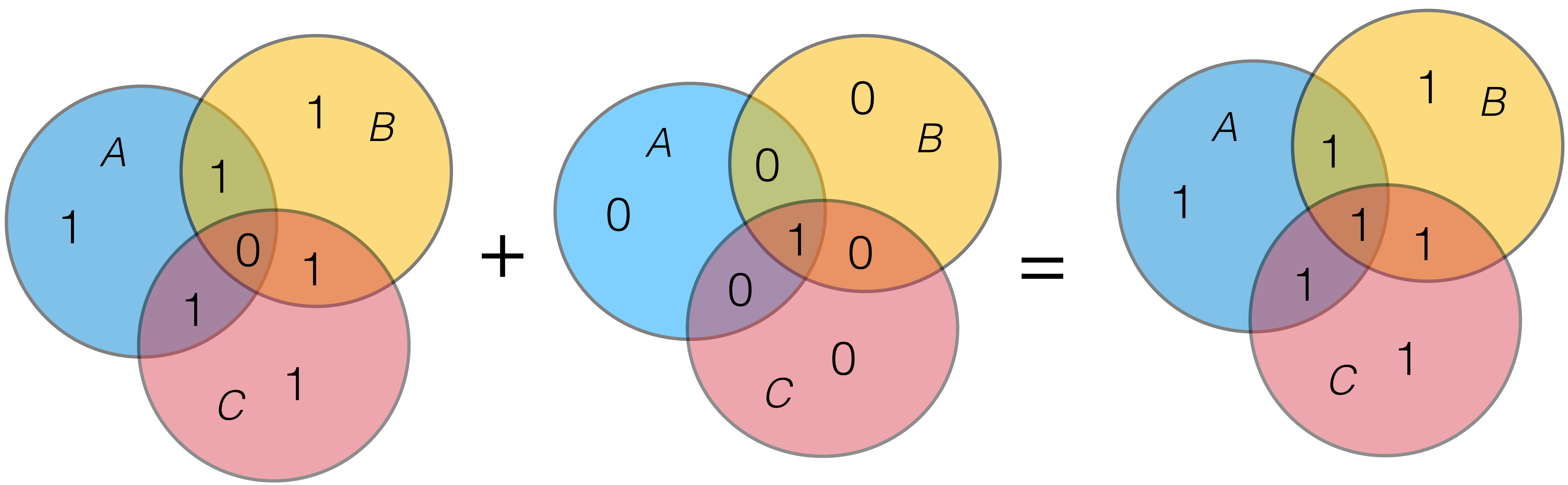
=



$|A| + |B| + |C| -$
$|A \cap B| - |B \cap C| - |A \cap C|$

Generalized Inclusion-Exclusion Principle

$$|A \cup B \cup C| =$$
$$|A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$



$|A| + |B| + |C| -$
$|A \cap B| - |B \cap C| - |A \cap C|$

$|A \cap B \cap C|$

$|A| + |B| + |C| -$
$|A \cap B| - |B \cap C| - |A \cap C|$
$+ |A \cap B \cap C|$

Generalized Inclusion-Exclusion Principle

Theorem (Generalized Inclusion-Exclusion Principle).

Let $A_1, \ldots, A_n$ be finite sets. Then

$$|A_1 \cup \ldots \cup A_n| = |A_1| + \ldots + |A_n| - \sum_{1 \le i < j \le n} |A_i \cap A_j|$$

$$+ \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| + \ldots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|.$$

Generalized Inclusion-Exclusion Principle

Proof.

We show that the right-hand side of the equation counts each element of $A_1 \cup \cdots \cup A_n$ exactly once.

Let $a \in A_1 \cup \cdots \cup A_n$.

Then there are $i_1, \ldots, i_k$: $1 \leq i_1 < \ldots < i_k \leq n$ such that $a \in A_{i_1} \cap \cdots \cap A_{i_k}$, moreover, for each $j \in \{1, \ldots, n\} \setminus \{i_1, \ldots, i_k\}$, $a \notin A_j$.

Then the number of times that a is counted in

$$|A_1| + \ldots + |A_n| - \sum_{1 \leq j_1 < j_2 \leq n} |A_{j_1} \cap A_{j_2}|$$

$$+ \sum_{1 \leq j_1 < j_2 < j_3 \leq n} |A_{j_1} \cap A_{j_2} \cap A_{j_3}| + \ldots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|$$

is $k - C(k, 2) + (-1)^4 C(k, 3) + \ldots + (-1)^{k+1} C(k, k)$

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is $k - C(k, 2) + (-1)^4 C(k, 3) + \ldots + (-1)^{k+1} C(k, k)$

$= 1 - (C(k,0) + (-1)^1 C(k, 1) + (-1)^2 C(k, 2) + (-1)^3 C(k, 3) + \ldots + (-1)^k C(k, k))$

$= 1 - (1-1)^k = 1.$

An Alternative Form of Generalized Inclusion-Exclusion Principle

Let N be the number of elements in the universe U.

Let A_i be the set of elements that have the property P_i.

$N(P_{i_1} \ldots P_{i_k})$:

The number of elements with the property $P_{i_1}, \ldots, P_{i_k}$.

Then $N(P_{i_1} \ldots P_{i_k}) = |A_{i_1} \cap \cdots \cap A_{i_k}|$.

$N(\overline{P_{i_1}} \cdots \overline{P_{i_k}})$:

The number of elements with none of the properties $P_{i_1}, \ldots, P_{i_k}$.

Then $N(\overline{P_{i_1}} \cdots \overline{P_{i_k}}) = N - |A_{i_1} \cup \cdots \cup A_{i_k}|$.

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$$N(\overline{P_1} \cdots \overline{P_n}) = N - N(P_1) - \ldots - N(P_n) + \sum_{1 \le i < j \le n} N(P_i P_j) - \sum_{1 \le i < j < k \le n} N(P_i P_j P_k) + \ldots + (-1)^n N(P_1 \ldots P_n).$$

Outline

- Generating functions: An introduction
- Useful facts about formal power series
  - Some basic facts
  - Extended binomial theorem
  - Some useful generating functions
- Solving recurrence relations by generating functions
- **Generalized Inclusion-exclusion principle**
  - Formulation and proof of the principle
  - **Applications**
    - **Sieves of Eratosthenes**
    - **Number of onto functions**
    - **Derangements**

The Sieves of Eratosthenes

The number of primes not exceeding 100.

By trial division theorem and fundamental arithmetic theorem,

$1 < x \leq 100$ is not a prime iff

it can be divided by a prime ≤ 10.

Primes ≤ 10: 2, 3, 5, 7

P_1: divisible by 2 P_2: divisible by 3

P_3: divisible by 5 P_4: divisible by 7

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P_1: divisible by 2 $\quad$ P_2: divisible by 3

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$N(P_1) = 50$ $\quad$ $N(P_2)$: $\lfloor 100/3 \rfloor = 33$ $\quad$ $N(P_3)$: $\lfloor 100/5 \rfloor = 20$ $\quad$ $N(P_4)$: $\lfloor 100/7 \rfloor = 14$

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$N(P_1P_2) = \lfloor 100/6 \rfloor = 16$ $\quad$ $N(P_1P_3) = \lfloor 100/10 \rfloor = 10$ $\quad$ $N(P_1P_4) = \lfloor 100/14 \rfloor = 7$

$N(P_2P_3) = \lfloor 100/15 \rfloor = 6$ $\quad$ $N(P_2P_4) = \lfloor 100/21 \rfloor = 4$ $\quad$ $N(P_3P_4) = \lfloor 100/35 \rfloor = 2$

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$N(P_2P_3P_4) = \lfloor 100/105 \rfloor = 0$ $\quad$ $N(P_1P_2P_3P_4) = \lfloor 100/210 \rfloor = 0$

The Sieves of Eratosthenes

The number of primes not exceeding 100.

$$N(\overline{P_1}\,\overline{P_2}\,\overline{P_3}\,\overline{P_4}) = N - N(P_1) - N(P_2) - N(P_3) - N(P_4) +$$
$$N(P_1P_2) + N(P_1P_3) + N(P_1P_4) + N(P_2P_3) + N(P_2P_4) + N(P_3P_4)$$
$$-\ N(P_1P_2P_3) - N(P_1P_2P_4) - N(P_1P_3P_4) - N(P_2P_3P_4) + N(P_1P_2P_3P_4)$$
$$= 99 - 50 - 33 - 20 - 14 +$$
$$16 + 10 + 7 + 6 + 4 + 2$$
$$- 3 - 2 - 1 - 0 + 0$$
$$= 21$$

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Number of Onto Functions

How many onto functions are there
from a set with 6 elements to a set with 3 elements ?

$A = \{a_1, \ldots, a_6\}$, $B = \{b_1, b_2, b_3\}$.

P_i (i=1,2,3): function f such that b_i is not in the range of f

Number of onto functions from A to B: $N(\overline{P_1}\,\overline{P_2}\,\overline{P_3})$

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N: 3^6

$N(P_1) = 2^6$, $N(P_2) = 2^6$, $N(P_3) = 2^6$

$N(P_1P_2) = 1$, $N(P_1P_3) = 1$, $N(P_2P_3) = 1$

$N(P_1P_2P_3) = 0$.

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$N(P_1) = 2^6$, $N(P_2) = 2^6$, $N(P_3) = 2^6$

$N(P_1P_2) = 1$, $N(P_1P_3) = 1$, $N(P_2P_3) = 1$

$N(P_1P_2P_3) = 0$.

$$N(\overline{P_1}\,\overline{P_2}\,\overline{P_3}) = N - N(P_1) - N(P_2) - N(P_3) + N(P_1P_2) + N(P_1P_3) + N(P_2P_3) - N(P_1P_2P_3) = 3^6 - 3 \cdot 2^6 + 1 + 1 + 1 - 0 = 540.$$

Number of Onto Functions

Theorem.

Let m and n be positive integers with $m \geq n$. Then, there are

$$n^m - C(n, 1)(n-1)^m + C(n, 2)(n-2)^m - \cdots + (-1)^{n-1}C(n, n-1) \cdot 1^m$$

onto functions from a set with m elements to a set with n elements.

$$\sum_{j=0}^{n-1} (-1)^j C(n, j)(n-j)^m$$

Number of Onto Functions

Proof of the theorem.

Suppose $A = \{a_1, \ldots, a_m\}$ and $B = \{b_1, \ldots, b_n\}$.

Let P_i be the property that b_i is not in the range of f: $A \to B$.

Then the number of onto functions is

$$N(\overline{P_1}\cdots\overline{P_n}) = N - \sum_{i=1}^{n} N(P_i) + \sum_{1\le i<j\le n} N(P_iP_j) - \ldots + (-1)^n N(P_1\cdots P_n).$$

Since $N(P_{i_1}\ldots P_{i_k}) = (n - k)^m$, we have

$$N(\overline{P_1}\cdots\overline{P_n}) =$$

$$n^m - C(n, 1)(n-1)^m + C(n, 2)(n-2)^m - \ldots + (-1)^{n-1}C(n, n-1)1^m.$$

Number of Onto Functions

Example.

How many ways to assign 5 jobs to 4 employees so that each employee is assigned at least one job ?

$m = 5$ and $n = 4$.

$$4^5 - C(4, 1)3^5 + C(4, 2)2^5 - C(4, 3)1^5$$
$$= 1024 - 4 \cdot 243 + 6 \cdot 32 - 4$$
$$= 240.$$

derangement: 错位排列

Derangements

A derangement is a permutation of objects that leaves no object in its original position.

derangement	non-derangement
12345	12345
$\downarrow$	$\downarrow$
21453	21543

D_n: the number of derangements of n objects.

e.g. $D_1 = 0$, $D_2 = 1$, $D_3 = 2$ ($123 \to 231$, $123 \to 312$).

Derangements

Theorem.

The number of derangements of a set with n elements is

$$D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right].$$

Derangements

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P_i: permutations $1 \cdots (i\text{-}1)\ i\ (i+1) \cdots n \rightarrow j_1 \cdots j_{i-1}\ i\ j_{i+1} \cdots j_n$.

derangement: 错位排列

Derangements

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$$N(\overline{P_1} \cdots \overline{P_n}) = N - \sum_{i=1}^{n} N(P_i) + \sum_{1 \le i < j \le n} N(P_i P_j) + \ldots + (-1)^n N(P_1 \ldots P_n)$$

$$= n! - C(n, 1)(n-1)! + C(n, 2)(n-2)! + \ldots + (-1)^n 0!$$

$$= \sum_{i=0}^{n} (-1)^i C(n, i)(n-i)! = \sum_{i=0}^{n} (-1)^i \frac{n!}{i!} = n! \sum_{i=0}^{n} (-1)^i \frac{1}{i!}.$$

Derangements

Corollary. $\lim_{n\to\infty} \frac{D_n}{n!} = \lim_{n\to\infty} \sum_{i=0}^{n} (-1)^i \frac{1}{i!} = e^{-1} \approx 0.368$

Derangements

Corollary. $\lim_{n\to\infty} \frac{D_n}{n!} = \lim_{n\to\infty} \sum_{i=0}^{n} (-1)^i \frac{1}{i!} = e^{-1} \approx 0.368$

for all x, $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$

$$e^{-1} = \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} = \lim_{n\to\infty} \sum_{i=0}^{n} (-1)^i \frac{1}{i!}$$

Recap

$G(x)$
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \cdots$
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \cdots$
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \cdots$
$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)x^k = 1 + C(n,1)x + C(n+1,2)x^2 + \cdots$

$G(x)$
$\frac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)(-1)^k x^k = 1 - C(n,1)x + C(n+1,2)x^2 - \cdots$
$\frac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)a^k x^k = 1 + C(n,1)ax + C(n+1,2)a^2x^2 + \cdots$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$

- Generating functions: An Introduction
- Useful facts about formal power series
  - Some basic facts
  - Extended binomial theorem
  - Some useful generating functions

Let $u \in \mathbb{R}$ and $k \in \mathbb{N}$. Then the *extended binomial coefficient*

$$\binom{u}{k} = \begin{cases} u(u-1)\cdots(u-k+1)/k! & \text{if } k > 0, \\ 1 & \text{if } k = 0. \end{cases}$$

Extended Binomial Theorem.
Let $x \in \mathbb{R}$ with $|x| < 1$ and $u \in \mathbb{R}$. Then

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k \qquad \text{Note: } 0^0 = 1$$

- Solving recurrence relations by generating functions

$$C_0 = 1,\ C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k} \qquad G(x) = \sum_{n=0}^{\infty} C_n x^n \qquad xG^2(x) - G(x) + 1 = 0 \qquad G(x) = \frac{1-\sqrt{1-4x}}{2x}$$

$$G(x) = \frac{1-\sqrt{1-4x}}{2x} = \frac{1}{2x} - \frac{1}{2x}\sum_{n=0}^{\infty} \binom{1/2}{n} (-4x)^n \qquad C_n = \frac{1}{n+1}\binom{2n}{n}$$

Recap

- Generalized inclusion-exclusion principle
  - Formulation and proof of the principle
  - Applications
    - Sieves of Eratosthenes
    - Number of onto functions
    - Derangements

Theorem (Generalized Inclusion-Exclusion Principle).

Let $A_1, \ldots, A_n$ be finite sets. Then

$$|A_1 \cup \ldots \cup A_n| = |A_1| + \ldots + |A_n| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| + \ldots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n| .$$

$$N(\overline{P_1} \cdots \overline{P_n}) = N - N(P_1) - \ldots - N(P_n) + \sum_{1 \le i < j \le n} N(P_i P_j) - \sum_{1 \le i < j < k \le n} N(P_i P_j P_k) + \ldots + (-1)^n N(P_1 \ldots P_n) .$$

Theorem.

Let m and n be positive integers with $m \geq n$. Then, there are

$$n^m - C(n,1)(n-1)^m + C(n,2)(n-2)^m - \cdots + (-1)^{n-1} C(n, n-1) \cdot 1^m$$

onto functions from a set with m elements to a set with n elements.

Theorem.

The number of derangements of a set with n elements is

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right].$$

Homework

1. Section 8.4.
   - Page 575, Exercise 8.
   - Page 577, Exercise 30, 36, 44.
2. Section 8.5-8.6.
   - Prove the generalized inclusion-exclusion principle by mathematical induction.
   - Page 591, Exercise 6.
   - Page 592, Exercise 24.

作业和答疑

1. 交作业时间：奇数周周一上课前（两周交一次）
2. 交作业方式（二选一）：
   - 通过https://mooc.ucas.edu.cn/portal课程网站
   - 上课现场提交
3. 答疑：助教崔乐天、刘易铖

Next

FINAL EXAM

基础软件验证研究室

如何用逻辑推理
保障基础软硬件
（芯片、操作系统、编译器等）的
正确性及安全性
研究室主页:
https://versys.ios.ac.cn/
欢迎大家后面来交流实习